LAB 4

AIM – Draw the Use Case Diagram for your project with main use cases as well as sub-use cases and a brief description

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Description -

1. System Overview

The Use Case Diagram models the functional requirements and user interactions within the MSME Marketplace system. It defines the system boundary and illustrates how distinct external actors interact with the platform's core modules.

2. Actors and Primary Responsibilities

The system identifies four primary actors with specific roles:

- **Seller:** An MSME producer who interacts with the system to "Manage Listing" by creating commodity. Sellers also "Track Order Status" and finalize deals through the "Negotiate & Place Order" use case.
- **Buyer:** This user interacts with the system to "Browse & Filter Catalog" to find available listings and uses "Post Buyer Requirement" to specify bulk purchase needs . Buyers also "Track Order Status" and engage in "Negotiate & Place Order".
- **Admin:** The platform administrator interacts with "Configure Vertical" to define structural rules like units of measurement, grading schemas, and price calculations. They also resolve conflicts via the "Manage Disputes" use case.
- **Verifier:** A specialized quality-assurance actor who executes "Review Flagged Grading" to manually inspect and confirm listings that receive low-confidence scores from the AI grading system.

3. System Relationships (Includes and Extends)

The model utilizes structural relationships to map workflows accurately:

- **Include Relationships :** These represent mandatory sub-processes required to complete a base use case.

  Manage Listing -> Grading Evidence: A seller is forced to upload supporting documents or photos when creating a listing.

  Manage Listing -> AI-Suggested Grade: The system automatically forces an AI estimation based on the uploaded evidence when a listing is created.

  Post Buyer Requirement -> Estimate Order Cost: When posting a requirement, the system automatically provides the buyer with an estimated total cost.

  Negotiate & Place Order -> Counter-Offer: Finalizing an order inherently requires the submission of a bid or counter-offer between the buyer and seller.

  Manage Disputes -> View Grading & Dispute Audit Trail: To resolve a dispute, the system mandates that the Admin inspect the retained history of grading decisions.

- **Extend Relationships :** These represent optional or conditional workflows triggered under specific system states.

  Review Flagged Grading -> Manage Listing: Human verification only triggers and extends the listing process if the AI-Suggested Grade returns a low confidence score.

  Allocate Order Across Multiple Sellers -> Post Buyer Requirement: This complex matching algorithm only triggers if a buyer's volume requirement cannot be fulfilled by a single seller's listing, splitting the order accordingly.

USE CASE DIAGRAM -

